IBM BOB HACKATHON · THEME 1: EXPLORE, FIX, AND BUILD

OnRamp

OnRamp makes IBM Bob prove its code changes, not just make them.

A drop-in `.bob/` kit for joining an unfamiliar codebase: record the baseline first, write a failing test before the fix, verify against the baseline, and ship evidence a reviewer can check.

22 → 0
errors from Bob's first fix, caught by Bob in Verifier mode and fixed

1,767
tests after the fix and feature, 0 regressions (independent run by the team)

491
Bob tool calls over 10 recorded tasks

85.9%
of the `.bob/` kit lines come from Bob commits (git blame)

Solo participant · Built with IBM Bob IDE 2.2.1 and Bob Shell 2.0.5
Code: github.com/KnigguKniggu-droid/onramp · Live demo: knigguKniggu-droid.github.io/onramp-evidence

THE DEVELOPER PROBLEM

The suite is red. Did you break it, or was it already broken?

29 / 1,761

Healthchecks tests fail on Windows before anyone changes a line.

Healthchecks is an open-source Django app. 26 tests need `socket.AF_UNIX`, 2 use a Linux-only date format and 1 needs time-zone switching. The team's Linux run passes all of them.

- Newcomers inherit large repositories, local quirks and test suites that already fail on their machine.
- AI agents make plausible changes fast. "It passes" is not evidence when the baseline is already red.
- Without a baseline, a reviewer must trust the agent or redo the work.

> **Our question:** can IBM Bob onboard onto unfamiliar code, fix a real bug and add a real feature, so that every claim can be checked?

WHAT WE BUILT

OnRamp: one folder you drop into a repository

STEP 1

/onramp

Validates the install, writes a repo profile, captures the baseline and maps the codebase.

STEP 2

Explorer mode

Read-only guide with file references. It answers "where does X live?" before any change.

STEP 3

Reproducer mode

Failing test first. The fix comes only after the pre-fix artefact shows `FAIL`.

STEP 4

Verifier mode

Re-runs the tests and classifies each failure against the baseline: inherited, regression or fixed.

STEP 5

Evidence report

Pre-fix failure, post-fix pass, diff and `REPORT_<slug>.md` for the reviewer.

Regression found in step 4 → back to the fix. No new failures → evidence report.

3

custom modes in `.bob/custom_modes.yaml`

4

skills: map-codebase, bug-to-test, verify-change, write-evidence-report

2

commands, including `/onramp`

1

PostToolUse hook in `.bob/settings.json`

Only Bob-native extension points. A team-written normalizer (`onramp_normalize.py`, 24 tests) is labelled separately.

PROOF · BLIND FIX OF A REAL UPSTREAM BUG

Bob broke 22 tests, caught it in Verifier mode, and fixed it

1 · SEALED

Symptoms only

Healthchecks at upstream `0968fa2`. Webhooks that answer HTTP 205 are retried 3 times and marked failed. The brief was SHA-256 sealed.

2 · REPRODUCER

Failing test first

`AssertionError: 3 != 1` requests, before any fix.

3 · FIRST FIX

22 errors

`200 <= status < 300` broke 22 tests in the 48-test webhook suite.

4 · VERIFIER

Caught and revised

Bob compared before and after, then changed the check to `range(200, 300)`. Webhook suite: 48 tests, OK.

5 · INDEPENDENT

0 regressions

The team's full run: 1,767 tests, same 29 inherited failures. A Linux re-run reproduced both results.

> "Before fix: 47 tests, OK. After fix: 48 tests (1 new), but 22 ERRORs. These 22 errors are regressions introduced by my fix!"
>
> IBM Bob, Verifier mode, step 40 · `docs/evidence/BOB_SESSION_webhook-205.md`

Then we revealed the maintainer's fix (`0a9bff9`)

Same defect, same line, different policy. Bob accepts every 2xx response; the maintainer added only 205. Bob's own report concludes the narrower upstream fix is safer, because `HttpTransport` is shared by about 25 integrations.

HOW IBM BOB WAS USED

Bob built the kit, and the kit made Bob prove its work

491
tool calls over 10 recorded tasks, exported from Bob's task database

23
commits with the `Generated-by: IBM Bob` trailer

85.9%
2,444 of 2,846 `.bob/` lines from Bob commits (git blame)

3 · 4 · 2 · 1
modes · skills · commands · hook, all Bob-native

- **Plan mode** designed the kit before building.
- **Agent mode** built the modes, skills, commands, hook and report generator, and wrote the bug fix and the feature.
- **Custom modes + `switch_mode`**: Reproducer and Verifier drove the blind fix.
- **Skills**: map-codebase wrote `CODEBASE_MAP.md` for Healthchecks.
- **Bob Shell (ACP)** ran the portability tests in a second repository.

Division of work

The team directed Bob, verified each result independently (full suites, revert checks, baseline diffs) and reported only observed failures back to Bob.

`docs/evidence/BOB_ACTIVITY_TOTALS.md`

PORTABILITY AND BUILD

Not hardcoded: same kit on a second app, plus a real feature

TaskFlow (Flask): the kit was copied in unchanged, three times.

RUN 1

Django assumptions

Hardcoded paths, output parsing and file names. Bob made the kit profile-driven.

›

RUN 2

One parser gap

`unittest` error types recorded as "Unknown" without `-v`. Bob fixed it in `8165552`.

›

RUN 3

Correct

Kit at `8165552`, copied unchanged. `AssertionError` classified correctly.

Feature #1090 (a user request)

JSON webhook bodies are pretty-printed in the ping details view. Non-JSON and email bodies are untouched, and Download Original still serves the exact bytes.

An independent check found `json.dumps` escaping "café". Bob added a test and fixed it.

While Bob had budget, every gap went back to Bob as an observed failure; later fixes are separate, team-labelled files. The 6 new tests in the 1,767-test run are Bob's webhook test and its 5 feature tests.

LIVE DEMO · TRY IT YOURSELF

Paste your own test log. Get a baseline-aware verdict.

knigguknigguk-droid.github.io/onramp-evidence

- **Open** the Evidence Explorer. It runs in the browser; nothing to install.
- **Drop in** a test log, or use the published Healthchecks runs.
- **Read** the verdict: every failure is classified against the baseline.
- **Check** the evidence files behind every claim in this deck.

inherited · regression · fixed · new passing

```
Verdict · SUITE_FULL_after_fix_independent.txt
Ran 1767 tests
FULL: 29 inherited · 0 fixed · 0 regressions
0 new failures · 6 new passing tests
1,732 unchanged passes
```

The team's independent run after Bob's fix and feature. The browser classifier is team-written; CI checks that it agrees with `onramp_normalize.py`.

IMPACT FOR TEAMS AND ENTERPRISES

Reviewers check evidence instead of trusting the agent

Faster onboarding

A newcomer runs `/onramp`. Explorer mode and the map-codebase skill map the repository read-only before anyone changes it.

Reviewable AI changes

Every change ships with a pre-fix failure, a post-fix pass, a diff and a report. A reviewer can re-run each claim.

Less test noise

The baseline separates failures that were already there from regressions the change introduced. 29 inherited failures stop blocking work.

Drop-in adoption

One `.bob/` folder and a per-repository profile. No new service to run.

Agent benchmarking

Sealed blind briefs compare an AI fix with the maintainer's real fix, on the same bug.

Audit trail

Every Bob commit carries `Generated-by: IBM Bob`. Bob's task log is exported as evidence.

ROADMAP AND LEARNINGS

Next: turn the rules into gates

1. **PreToolUse gate**: block production edits until a failing-test artefact exists.
   Today the mode rules are instructions. When Verifier refused a production edit, Bob switched itself to Agent mode.

2. **Fail-closed verifier**: a crashed or partial run must fail, not pass silently.
   The team-written `onramp_normalize.py classify` already fails closed.

3. **pytest support**: profiles for pytest projects, then Node and Java.

4. **CI and benchmarks**: classify every pull request; add blind benchmarks from real upstream fixes.

Known limitations: see README.

Learnings

- A baseline is the difference between "tests are red" and "I broke something".
- A narrow check misses regressions. Bob's first check ran one app; Verifier's wider run found the 22 errors.
- Sealing the brief makes an AI fix comparable with a human one.
- A second repository exposes what a single repository hides.

CHECK EVERY CLAIM

Links and evidence

Live demo	knigguknigguk-droid.github.io/onramp-evidence
Code and kit	github.com/KnigguKniggu-droid/onramp
Bob's blind-fix session	docs/evidence/BOB_SESSION_webhook-205.md (search "22 ERRORs")
Bob activity totals	docs/evidence/BOB_ACTIVITY_TOTALS.md
Linux reproduction	docs/evidence/LINUX_REPRODUCTION_2026-10-19.md

1,761 / 29
baseline tests / inherited failures

22 → 0
errors in the 48-test webhook suite

1,767 / 0
tests / regressions (team's run)

491 / 23
Bob tool calls / Bob commits

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